# Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia

## Week 7 — Peer review: how it works

Francis Bond (**Academic**) Pavel Bednařík (Wiki)

Palacký University Olomouc | Wikimedia ČR

# Today

1. Feedback and Revision
2. Peer Review, Trust, and Metrics
3. Problems: Bias & Inequality, (un)Reproducibility

# Roadmap

1. Feedback and Revision
2. Peer Review, Trust, and Metrics
3. Problems: Bias & Inequality, (un)Reproducibility

# Why feedback?

- First drafts are rarely clear, complete, or well-argued
- Feedback (from peers, tutors, self-reflection) helps identify:
  - unclear claims
  - logical gaps
  - missing evidence
  - stylistic issues
- Better writing is clearer and more persuasive, and when you write about people and their data, more responsible

# What feedback gives you

- As the writer:
  - Another reader's view: it shows your assumptions and blind spots
  - Fresh eyes on the structure: does the argument build?
  - A chance to add evidence and answer counter-arguments
  - A chance to fix style and tone for the audience
- As the reader:
  - You learn to weigh evidence and spot weak arguments
  - You see how other people solve the problems you have
  - Better writing, better discussion
- Think of the review you wrote on Moodle: which of these did it do?

# Feedback as an ethical check

- Writing about people, cultures and data carries responsibility: a reader can catch bad framing, bias or harm you did not see
- Facts get checked, sources verified, sensitive topics handled with care
- Careful writing builds trust in scholarship as a whole

# Roadmap

1. Feedback and Revision

2. Peer Review, Trust, and Metrics
   - Feedback Beyond Academia
   - Reviewing at the Association for Computational Linguistics

3. Problems: Bias & Inequality, (un)Reproducibility

# What peer review is

- Experts evaluate a manuscript before it is published
- Its aims are to:
  - Check the **soundness** of methods and argument
  - Improve clarity, structure, and contribution
  - Build **trust** in published research
- Reviewers are gatekeepers and, at best, mentors

# Types of peer review

- Three models, trading off transparency, fairness and accountability

**Single-blind:** Reviewers know the author's identity, but authors do not know the reviewers

- Common in science and social science
- Advantage: reviewers can assess expertise and potential conflicts
- Risk: unconscious bias (e.g., by gender, affiliation, language, or region)

**Double-blind:** Neither authors nor reviewers know each other's identity

- Reduces bias (see the WSDM experiment later)
- Often used in humanities and linguistics conferences
- Risk: less accountability, and reviewers can often guess the authors

**Open review:** Both sides know identities, and sometimes reviews are published

- Transparent, and reviewers get credit for their work
- Politer, but reviewers may hold back (see the BMJ trial later)
- Used in some open-science and data journals

# Peer review: strengths and weaknesses

**Strengths**

- Catches errors before publication
- Provides expert feedback before publication
- A shared standard: “peer reviewed” means something
- Dialogue within a discipline

**Weaknesses**

- Can be slow, inconsistent, or biased
- Overburdens a small reviewer pool
- Reviewers rarely check replication
- Negative or null results often unpublished (the file-drawer effect)

# The third reviewer

- Every academic has a story about the reviewer who sank their paper
  - Scientific Peer Review, ca. 1945: a *Downfall* parody — two good reviews, then the third reviewer wants more experiments
  - Shit My Reviewers Say: real review comments, collected
- The joke has since moved to **Reviewer 2**
- Peterson (2020) tested it on the reviews of one journal (*Political Behavior*):
  - Reviewer 2 is no more negative than the others
  - Reviewer 3 is the one most often far below the other reviewers
- The point for you: a harsh review is one reader's view — read it for what you can fix

# Evolving models of peer review

- **Post-publication review:**
  - Articles or datasets are published first, then openly discussed
  - Examples: F1000Research, PubPeer
- **Community review:**
  - Large-scale collaborative commenting before publication (e.g., open preprints, GitHub issues, Open Review at LangSci Press)
  - Many eyes, in the open
- **Preprint feedback:**
  - Early sharing (e.g., arXiv, LingBuzz) gets informal feedback before submission
  - Faster, and it establishes priority (no fear of being scooped)
- In all three, review is a conversation that continues after publication

# Feedback in industry: giving notes

- In film, publishing, games and advertising, feedback is called **notes** (Catmull & Wallace, 2014)
- Notes come from editors, producers, managers and clients, usually down a hierarchy
- The goal is the same as in academic review:
  - make the work better
  - make sure it does its job for its audience
- The constraints differ:
  - deadlines, budgets, brand guidelines, lawyers
  - the writer, the editor and the client may want different things

Video: Ed Catmull on the Pixar Braintrust (Stanford, 8 min)

# How industry notes differ from academic peer review

- **Hierarchical**: the notes come from someone with authority over the project
- **Goal-driven**: the note serves the project, brand or product
- **Negotiated, up to a point**: creators argue back, but the final call may not be theirs
- **Iterative**: notes, revisions, more notes, approval
- **Success** means the work does what the project needs

# Best practices from industry feedback culture

- **Clarity first**: say what the problem is before proposing a fix (the Pixar rule)
- **Respect the creator's intent**: improve the work without imposing your style
- **Prioritise**: not every note can be taken; choose the ones that serve the project
- **Negotiate**: creators can ask what a note means, and push back
- **Shared responsibility**: notes name problems; the creator chooses the fix

# What students can learn from industry feedback

- You will get feedback from editors, managers, clients and colleagues
- What carries over:
  - receiving feedback without defensiveness
  - interpreting conflicting goals
  - deciding which suggestions to adopt, adapt, or reject
  - communicating clearly about revisions

# Feedback cultures beyond academic peer review

- **Journalism**: editors and fact-checkers check claims against primary sources before publication (academic reviewers rarely check that a citation says what the paper claims)
- **Wikipedia**: articles pass through community review; Good and Featured Article criteria require verifiability and a neutral point of view
- **Code review**: peers review code for bugs, style and maintainability, in public and tied to version control (GitHub pull requests)

# Peer review in Computational Linguistics

- Whole communities invest in improving peer review
- The Association for Computational Linguistics (ACL) publishes:
  - reviewer training guides (ACL 2023 Program Committee, 2023)
  - best-practice recommendations (ACL Rolling Review, 2025)
  - annual reports analysing what works and what fails (ACL Standing Committee on Peer Review, 2025)
- A large conference at scale — ACL 2025, for example:
  - 8,360 submissions
  - 1,699 accepted to the main conference (20.3%), 1,392 as Findings (16.7%)
  - 169 Senior Area Chairs, 1,937 Area Chairs, 11,720 reviewers!
  - Each paper reviewed by three reviewers, the authors respond, the reviewers update their reviews, the area chairs suggest which papers are accepted, the senior area chairs make the final decisions, …

# Accept reviews only when appropriate (ACL guidelines)

- Before agreeing to review, check:
  - **Expertise**: Can you fairly assess the methods and claims?
  - **Conflicts of interest**: shared institution, collaborators, competitors
  - **Capacity**: do you have time to read it carefully?
- Accepting when you are unqualified or overloaded leads to unfair or rushed reviews
- Slightly undermined by the fact that if you submit a paper, someone from the author team must volunteer to review, and owes four reviews per submission (capped at 16 in a cycle), …
  - ACL 2025 accepted 20.3% to the main conference and a further 16.7% as Findings — four papers miss the main conference for every one that makes it, and nearly two thirds are turned down altogether
  - From October 2026 the ACL sustainable reviewing policy goes further:
    - each submission needs a qualified service contributor (at most two submissions each)
    - authors are capped at 20 submissions and 5 first-author submissions a cycle
    - no registered service, no review: the paper is desk rejected

# Read carefully and generously (ACL guidelines)

- First read: understand the research question and contribution
- Second read: evaluate evidence, structure, and clarity
- Look for what the author is trying to do
- Avoid snap judgments based on style, formatting, or first impressions

# Write constructive, actionable feedback (ACL guidelines)

- Every review should include both:
  - **Strengths**: what works
  - **Weaknesses**: what to fix
- Make criticisms specific:
  - point to a sentence or section
  - explain why it is unclear
  - suggest how to fix it
- Vague comments (“the paper is confusing”) do not help

# Check evidence, ethics, and reproducibility (ACL guidelines)

- A reviewer is responsible for more than novelty
- Consider:
  - Are claims supported by evidence?
  - Are methods and data described clearly?
  - Are ethical implications acknowledged?
  - Can results be verified or reproduced?

# Professionalism and respect (ACL guidelines)

- Use a constructive and collegial tone
- Avoid dismissive language:
  - “This is trivial”
  - “obviously wrong”
  - “the authors don’t understand”
- Remember: reviewers and authors are colleagues
- Treat the paper the way you would want yours to be treated

# Reviewer power and responsibility (ACL COPR)

- Reviewers gain **access to unpublished work** before the community
- This creates responsibility:
  - Never share, quote or use ideas from the manuscript
  - Never delay a review to gain priority
  - Never punish papers that challenge your favourite theory
  - Declare conflicts of interest
- Rockwell (2005): access to unpublished work is a power imbalance, and reviewers must not exploit it
- Peer review runs on trust

# What this means for you (student reviewers)

- The same principles apply in the classroom:
  - read generously
  - comment specifically
  - focus on clarity, evidence, and logic
  - be respectful and supportive
- Practising good peer review improves your own writing:
  - you learn what makes an argument work
  - you see how readers interpret drafts
  - you build editorial judgment

Based on ACL reviewer training materials (ACL 2023 Program Committee, 2023; ACL Rolling Review, 2025; ACL Standing Committee on Peer Review, 2025).

# ACL-inspired peer review checklist

- Am I qualified and conflict-free?
- Have I read the paper carefully (twice)?
- Did I identify strengths and weaknesses?
- Are my comments specific and actionable?
- Did I evaluate ethics, evidence, and reproducibility?
- Is my tone respectful and professional?
- Have I upheld confidentiality and integrity?

# Roadmap

1. Feedback and Revision
2. Peer Review, Trust, and Metrics
3. Problems: Bias & Inequality, (un)Reproducibility
   - Punktoza
   - Bias and Inequality in Peer Review
   - The Reproducibility Crisis
   - AI and Peer Review

# Publish or Perish

- Academics are under pressure to publish
- This is a major part of how they are evaluated
- This should lead to more, higher quality publications
- But can lead to many more, lower quality publications
  - Or even fraud!
- And means we have to review more and more publications

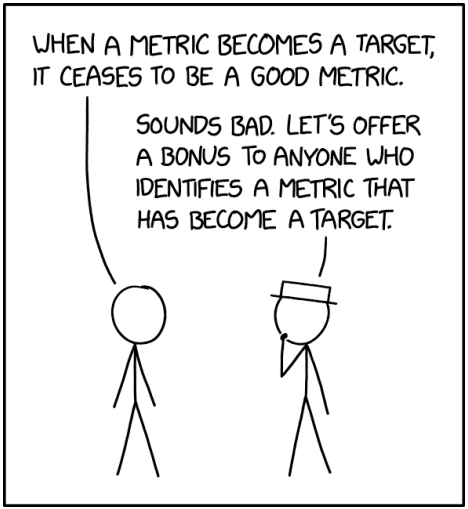
# Metrics: why they were introduced

- Evaluators wanted standard numbers to compare:
  - journals (Impact Factor)
  - researchers (h-index)
  - institutions (citation counts, productivity)
- **Impact Factor (IF):** citations made in one year to what a journal published in the *two preceding* years, divided by the number of “citable items” (research articles and reviews) it published in those two years
  - Designed to help libraries choose which journals to buy
  - A standard criticism is the mismatch between the two halves: the numerator counts citations to *everything* (editorials, letters, news), the denominator counts only the “citable items”
- **h-index:** a researcher has index *h* if they have *h* papers each cited at least *h* times
  - Productivity and influence in one number
- Meant to inform expert judgement, not to replace it

# Metrics: when measures become targets

- When hiring, promotion, or funding is tied to metrics:
  - Researchers optimise for what is counted, not what is meaningful
  - Citation counts become currency; novelty outranks replication or rigour
- **Goodhart's Law**, as Strathern (1997, p. 308) put it:
  - "When a measure becomes a target, it ceases to be a good measure"
  - IF and h-index stop measuring anything once people game them
- **Punktoza** ("points fever"): publishing to collect evaluation points (Kulczycki, 2017a, 2017b)
  - Encourages salami-slicing, low-risk incremental work, quantity over substance
  - A rational response to points-based national evaluation

# Goodhart's Law xkcd 2899



# Peer review is not neutral

- Peer review should judge the work alone, but social biases shape outcomes
- Studies find bias by gender, prestige and affiliation (next two frames)
- The result: the same people stay visible

# Gender bias and underrepresentation

- A study of 145 journals (1.7 million authors and 740,000 referees) found that manuscripts by women were not universally penalized, yet women remain under-represented among reviewers and editors (Squazzoni et al., 2021)
- An earlier study of 9,000 editors and 43,000 reviewers at one publisher found the same under-representation, and that reviewers tend to be matched with authors like themselves (Helmer et al., 2017; Murray et al., 2018)
- In grant-review contexts, women have also been shown to receive lower scores despite similar qualifications (Wennerås & Wold, 1997)

# Affiliation and prestige bias

- Affiliation bias has been measured directly: WSDM 2017 ran a controlled experiment in which every paper got two *single-blind* reviewers and two *double-blind* ones (Tomkins et al., 2017)
  - The single-blind reviewers were markedly more likely to recommend acceptance for papers by famous authors (odds $\times 1.63$) and from top universities ($\times 1.58$) or top companies ($\times 2.10$)
  - Their double-blind colleagues, reading the same papers, showed no such advantage — so double-blinding takes it away
- So publication records amplify institutional inequality: the rich get richer

# Open review — which is what you just did

- Your reviews went in the Moodle forum: **signed**, and readable by the whole class. That is **open review**, and some journals work this way (eLife, BMJ, F1000).
- **What blinding buys you**: double-blinding removes the famous-author advantage (Tomkins et al., 2017).
- **What signing costs you**: in a randomised trial at the *BMJ*, reviewers asked to be identified to the author were **12 points more likely to refuse** to review at all (35% v 23%) — but for those who did review, quality, recommendation and time taken were **no different** (van Rooyen et al., 1999).

**So, from your own week:**

- Did you soften anything because your name was on it?
- Did reading other people's reviews change how you wrote yours?
- Would you have said more, or less, anonymously?

# Why these biases happen, and what we can do

- Why these biases happen
  - Reviewers and editors lean on status cues: institution, reputation, familiarity
  - Reviewing is human work, and unconscious bias (gender, prestige, nationality, career stage) is common
  - It is structural too: who is asked to review, whose papers get seen, who gets accepted
- What this means for us
  - As reviewers: judge the content, and notice when you are judging the name
  - As authors: a rejection is not only about quality
  - As a community: diverse reviewers and editors, anonymised review where it fits, transparent editorial decisions

# What is reproducibility / replicability?

- **Reproducibility:** re-analyse the same data/code to get the same result
- **Replicability:** collect new data under same conditions and test whether the effect holds
- Without either, a published finding is provisional (Goodman et al., 2016)

# The crisis in numbers

- In a 2015 project replicating 100 psychology studies, only about 36% of results were successfully replicated; effect sizes dropped by about 50% (Open Science Collaboration, 2015)
- A 2016 *Nature* survey of 1,576 researchers, across all the sciences, found over 70% had tried and failed to reproduce another scientist's results, and 52% judged that there is a significant crisis (Baker, 2016)
- Cancer biology is no better: an eight-year project repeated 50 experiments from 23 high-profile papers, and only 46% of the replications (51/112) succeeded, with effect sizes a median 85% smaller (Errington et al., 2021)
- If findings routinely fail to replicate, the literature misleads everyone who builds on it

Video: Veritasium: "Is Most Published Research Wrong?" (12 min)

# Causes: why reproducibility fails

- Low statistical power (too few subjects): effects that come and go (Ioannidis, 2005)
- Publication bias: positive, novel results get published; null results stay in the drawer
- Missing code, data and method details: honest researchers make replication impossible by not sharing
- Incentives: rewards go to novelty, impact and speed, none to replication

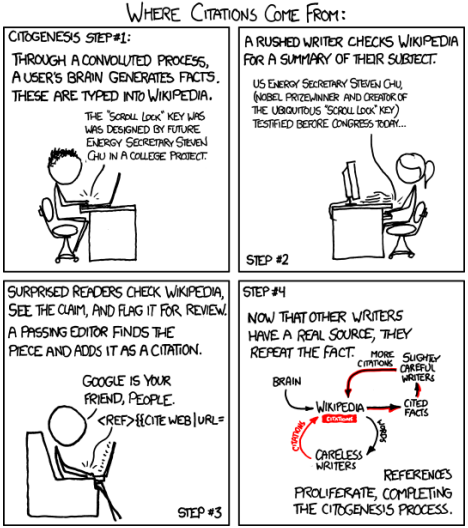
# FAIR, CARE and good writing help

- FAIR: metadata, shared code and data make re-analysis possible
- CARE: honest reporting of context and limitations
- Good writing: a reader can tell exactly what was done
- Together: more reliable science

# Citation as trust

- A citation says: I read this and trust it enough to build on it
- Citations add up to networks of credibility
- Not all citations are equal:
  - Some endorse, others critique or merely mention
  - Citation cascades pass on claims no one checked (next slide)
- Cite what you have read, and say what you take from it

# Citogenesis xkcd 978



# AI in peer review: a known problem

- Large language models (LLMs) are used in academic writing, and in peer review
- A 2024 study estimated that 6.5–17% of peer reviews at major AI conferences may have been substantially written or modified by LLMs (Liang et al., 2024)
- The tell: *commendable*, *meticulous* and *intricate* rose 10–35 times in frequency
- Major publishers all restrict AI in peer review, but not in the same way:
  - **All of them:** never upload the manuscript, or any part of it, to a public AI tool: that alone breaks confidentiality
  - **Taylor & Francis** and **Sage:** do not let AI generate the review at all (T&F allow it only to improve the language of a review you wrote yourself)
  - **Springer Nature:** AI may support a reviewer but never replace their judgement; use only secure or institutionally approved tools, and be transparent about it

# Why AI-generated reviews are problematic

- **Confidentiality breach**: uploading a manuscript to an LLM shares it with a third party
- **Accountability**: reviewers answer for accuracy and judgement, and an LLM cannot
- **Bias and hallucination**: LLMs repeat biases and produce plausible, wrong critiques
- **Superficiality**: generated reviews are generic and miss the issues an expert would see
- **Trust erosion**: an automated review is no longer an expert's judgement

# Typical guidance on AI in peer review

- Most journals and conferences now require:
  - Disclosure of any AI assistance in writing or reviewing
  - Human accountability for all submitted content
  - No uploading of confidential manuscripts to AI tools
- Acceptable uses (with disclosure):
  - Grammar and spell-checking
  - Improving clarity of your own writing
- Unacceptable uses:
  - Generating review text
  - Summarising or analysing manuscripts you are reviewing
  - Submitting AI-generated content as your own expert judgment

# Acknowledgements

- I have had over 200 papers reviewed (thanks reviewers!)
- I have peer reviewed over 2,000 conference papers, 100 journal papers, 10 PhDs, 2 Habilitations, and many grants
- **GPT-4o** in ChatGPT (OpenAI, 2025) was used to format the references, and generate a first draft of the slides
- **Claude Opus 5** (Anthropic, 2026) was used to check the slides: to resolve every DOI against Crossref, fetch every URL, and compare each quotation with its original. The prompt was of the form

  ```
  Verify every claim, DOI, URL and quotation against a real source. Where you cannot
  verify something, say so rather than guessing --- never invent a reference.
  ```

  Each correction names the source it was checked against. Responsibility for what remains is mine.

# References I


- ACL 2023 Program Committee. (2023). Review basics: Guidance for reviewers (acl 2023) [Accessed 2025-03-10]. https://2023.aclweb.org/blog/review-basics/
- ACL Rolling Review. (2025). Reviewer guidelines [Accessed 2025-03-10]. https://aclrollingreview.org/reviewerguidelines
- ACL Standing Committee on Peer Review. (2025). Report 1 of the ACL standing committee on peer review [Accessed 2025-03-10]. https://www.aclweb.org/adminwiki/images/9/9e/COPR-5-cycle-report-to-publish.pdf
- Anthropic. (2026). *Claude Opus 5 (Sep 20, 2026 version)* (Large language model). Retrieved September 20, 2026, from https://claude.ai/
- Baker, M. (2016). 1,500 scientists lift the lid on reproducibility. *Nature*, *533*, 452–454. https://doi.org/10.1038/533452a
- Catmull, E., & Wallace, A. (2014). *Creativity, inc.: Overcoming the unseen forces that stand in the way of true inspiration* [Contains Pixar's principles on feedback, candor, and iterative notes sessions]. Random House.

# References II


- Errington, T. M., et al. (2021). Investigating the replicability of preclinical cancer biology. *eLife*. https://doi.org/10.7554/eLife.71601
- Goodman, S. N., Fanelli, D., & Ioannidis, J. P. A. (2016). What does research reproducibility mean? *Science Translational Medicine*, *8*(341), 341ps12. https://doi.org/10.1126/scitranslmed.aaf5027
- Helmer, M., Schottdorf, M., Neef, A., & Battaglia, D. (2017). Gender bias in scholarly peer review. *eLife*, *6*, e21718. https://doi.org/10.7554/eLife.21718
- Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLOS Medicine*, *2*(8), e124. https://doi.org/10.1371/journal.pmed.0020124
- Kulczycki, E. (2017a). Assessing publications through a bibliometric indicator: The case of comprehensive evaluation of scientific units in Poland. *Research Evaluation*, *26*(1), 41–52. https://doi.org/10.1093/reseval/rvw023
- Kulczycki, E. (2017b). Punktoza jako strategia w grze parametrycznej w Polsce. *Nauka i Szkolnictwo Wyższe*, *49*(1), 63–78. https://doi.org/10.14746/nisw.2017.1.4

# References III


- Liang, W., et al. (2024). Monitoring AI-modified content at scale: A case study on the impact of ChatGPT on AI conference peer reviews. *International Conference on Machine Learning*. https://doi.org/10.48550/arXiv.2403.07183
- Murray, D., Siler, K., Larivière, V., Chan, W. M., Collings, A. M., Raymond, J., & Sugimoto, C. R. (2018). Author-reviewer homophily in peer review. *bioRxiv*. https://doi.org/10.1101/400515
- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, *349*(6251), aac4716. https://doi.org/10.1126/science.aac4716
- OpenAI. (2025). *ChatGPT (Mar 14, 2025 version)* (Large language model). Retrieved October 14, 2025, from https://chat.openai.com/
- Peterson, D. A. M. (2020). Dear Reviewer 2: Go F' yourself. *Social Science Quarterly*, *101*(4), 1648–1652. https://doi.org/10.1111/ssqu.12824
- Rockwell, S. (2005). Ethics of peer review: A guide for manuscript reviewers [Accessed 2025-12-06].

# References IV


- Squazzoni, F., Bravo, G., Farjam, M., Marusic, A., Mehmani, B., Willis, M., Birukou, A., Dondio, P., & Grimaldo, F. (2021). Peer review and gender bias: A study on 145 scholarly journals. *Science Advances*, *7*(2), eabd0299. https://doi.org/10.1126/sciadv.abd0299
- Strathern, M. (1997). 'improving ratings': Audit in the British University system. *European Review*, *5*(3), 305–321. https://doi.org/10.1002/(SICI)1234-981X(199707)5:3<305::AID-EURO184>3.0.CO;2-4
- Tomkins, A., Zhang, M., & Heavlin, W. D. (2017). Reviewer bias in single- versus double-blind peer review. *Proceedings of the National Academy of Sciences*, *114*(48), 12708–12713. https://doi.org/10.1073/pnas.1707323114
- van Rooyen, S., Godlee, F., Evans, S., Black, N., & Smith, R. (1999). Effect of open peer review on quality of reviews and on reviewers' recommendations: A randomised trial. *BMJ*, *318*(7175), 23–27. https://doi.org/10.1136/bmj.318.7175.23

# References V

Wennerås, C., & Wold, A. (1997). Nepotism and sexism in peer-review. *Nature*, *387*(6631), 341–343. https://doi.org/10.1038/387341a0